

3 *Staphylococcus epidermidis* is a species of bacterium that lives on human skin.

Staphylococcus aureus is a pathogenic bacterium that can infect humans.

(a) *S. epidermidis* and *S. aureus* are prokaryotes.

Table 3.1 shows some cell features that could apply to typical prokaryotic cells or to typical eukaryotic cells or to both types of cell.

Complete Table 3.1 by using a tick (✓) if the feature applies to the type of cell or a cross (✗) if the feature does not apply to the type of cell.

Put a tick (✓) or a cross (✗) in every box.

Table 3.1

feature	prokaryotic cell	eukaryotic cell
circular DNA		
80S ribosomes		
a cell diameter of 20 μm		

[3]

(b) One way that *S. aureus* can infect humans is through wounds (breaks) in the skin.

Populations of *S. aureus* develop on human skin as part of a biofilm. The biofilm contains cells of *S. aureus* within a mixture of polymers that have been secreted by the cells.

S. epidermidis produces a protease enzyme that prevents the growth of *S. aureus* populations on human skin. Proteases catalyse the breakdown of proteins.

Fig. 3.1 is a diagram showing populations of *S. epidermidis* and *S. aureus* on human skin cells.

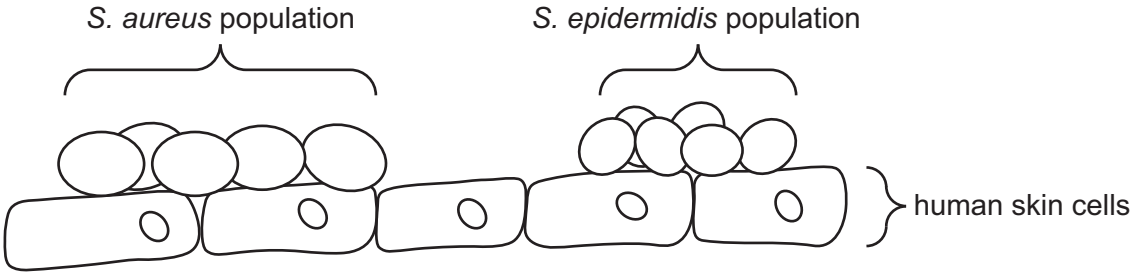


Fig. 3.1



Suggest **and** explain how the protease produced by *S. epidermidis* cells prevents the growth of an *S. aureus* population on human skin.

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(c) *S. aureus* can infect many tissues in the human body, including tissues in the gas exchange system.

(i) Describe the role of goblet cells in the protection of tissues in the trachea from infection by *S. aureus*.

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..... [2]

(ii) *S. aureus* cells can infect tissues by passing in between cells that line the lumen of the trachea.

State the name of a cell type, **other than** goblet cells, that lines the lumen of the trachea.

..... [1]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(ii) Some strains of *S. aureus* are resistant to vancomycin and penicillin.

Describe the steps that can be taken to reduce the impact of antibiotic resistance.

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[Total: 15]